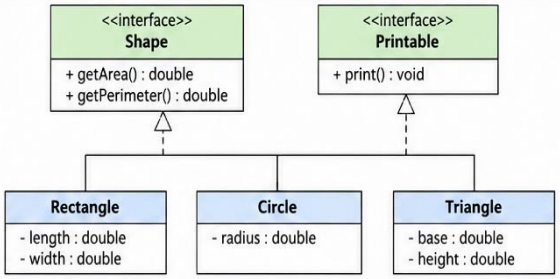
Q1. Shape and Printable (Interfaces with multiple inheritance) **[5 Marks]**

Write a Java program using the following class diagram. There are two interfaces named **Shape** and **Printable**. The interfaces have the methods as shown in the diagram. Create three classes **Rectangle**, **Circle**, and **Triangle** which implement both interfaces. Each class should have a constructor that takes the required values as arguments and must display the shape information along with a printable message. In the main class, create objects of each class and call all the methods.

Sample Output:
Rectangle – Area: 50.0, Perimeter: 30.0, Printable: Rectangle
Circle – Area: 78.54, Perimeter: 31.42, Printable: Circle
Triangle – Area: 24.0, Perimeter: 20.0, Printable: Triangle




```
interface Shape {
    double getArea();
}

interface Printable {
    void print();
}

class Rectangle implements Shape, Printable {
    double length;
    double width;

    Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }

    public double getArea() {
        return length * width;
    }

    public void print() {
        System.out.println("Shape: Rectangle");
    }
}
```

```
class Circle implements Shape, Printable {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }

    public double getArea() {
        return 3.14 * radius * radius;
    }

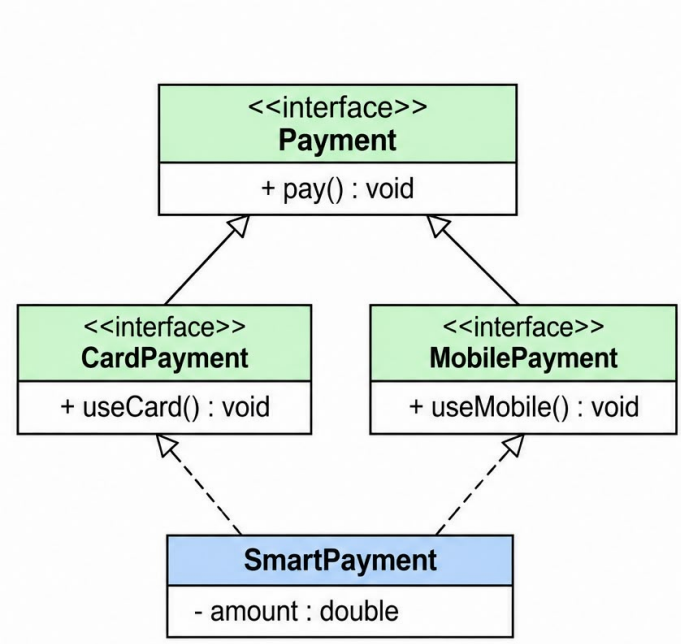
    public void print() {
        System.out.println("Shape: Circle");
    }
}

public class Main {
    public static void main(String[] args) {

        Rectangle r = new Rectangle(10, 5);
        Circle c = new Circle(5);

        r.print();
        System.out.println("Area: " + r.getArea());

        c.print();
        System.out.println("Area: " + c.getArea());
    }
}
```




```java
interface Payment {
    void pay();
}
interface CardPayment extends Payment {
    void useCard();
}
interface MobilePayment extends Payment {
    void useMobile();
}
class  SmartPayment implements CardPayment,
                                MobilePayment {
    double amount;

    SmartPayment(double amount) {
        this.amount = amount;
    }

    public void pay() {
        System.out.println("Payment of " + amount + " Tk"
                        + " is successful.");
    }
    public void useCard() {
        System.out.println("Paid using Card.");
    }
    public void useMobile() {
        System.out.println("Paid using Mobile.");
    }
}

public class Main {
    public static void main(String[] args) {
        SmartPayment sp = new  SmartPayment(1000.0);
        sp.pay();
        sp.useCard();
        sp.useMobile();
    }
}
```

Sample Output:
Payment of 1000.0 Tk is successful.
Paid using Card.
Paid using Mobile.

